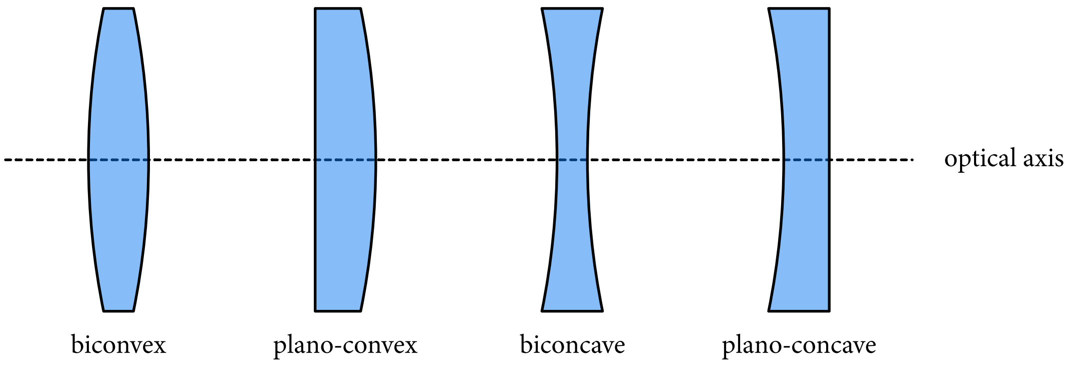
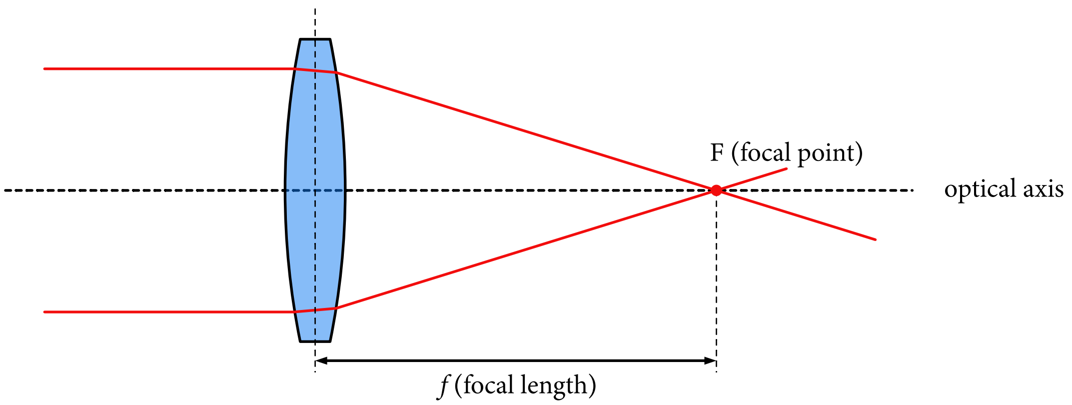
Lenses

A lens is a device made from a transparent medium (e.g. glass) that produces converging or diverging light rays due to refraction. Most lenses are spherical lenses, i.e. their surfaces are parts of the surface of spheres. Some common types of lenses are displayed below.

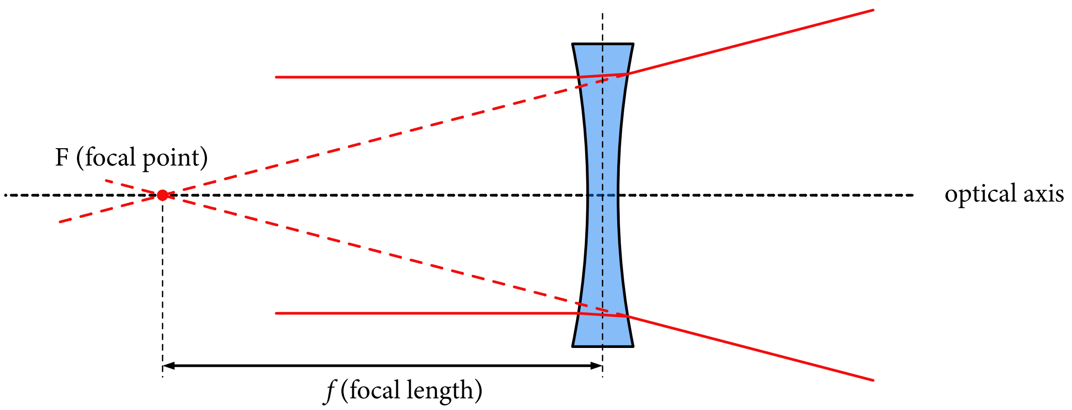


The *optical axis* is a line that passes through the centre of the lens and is perpendicular to its symmetry axis.

For convex lenses, parallel light rays entering the lens parallel to the optical axis converge (approximately) in a spot (*focal point* or *focus*). The lens is called a *positive* or *converging* lens. A thin convex lens‡ is characterised by its *focal length*, i.e. the distance between the centre of the lens and the focal point.



For concave lenses, parallel light rays passing through the lens spread (diverge). The lens is called a *negative* or *diverging* lens. After passing through the lens, they appear to come from a focal point in front of the lens. The (negative) distance from this point to the centre of a thin concave lens is called focal length.



‡ For a thick lens, the focal length is the distance to one of two *principal planes*. For a thin lens, the principal planes can be considered to be at the same position.

Lensmaker's Equation

The focal length for a spherical length is given by the lensmaker's equation:

$$\frac{1}{f} = (n-1)\left[\frac{1}{R_1} + \frac{1}{R_2} - \frac{(n-1)d}{n R_1 R_2}\right]$$

Where n is the refractive index of the lens material, R_1 and R_2 are the radii of curvature of the two surfaces (positive for convex and negative for concave), and d is the thickness of the lens (measured on the optical axis).

For a thin lens ($d \ll R_1, R_2$), the lensmaker's equation can be approximated as

$$\frac{1}{f} \approx (n-1)\left[\frac{1}{R_1} + \frac{1}{R_2}\right]$$

Examples:

- Symmetrical biconvex lens ($R_1 = R_2 = R$):

  $$\frac{1}{f} \approx (n-1)\left[\frac{1}{R} + \frac{1}{R}\right] = \frac{2(n-1)}{R} \Longrightarrow f = \frac{R}{2(n-1)}$$

- Plano-convex lens ($R_1 = R, R_2 = \infty$):

  $$\frac{1}{f} \approx (n-1)\left[\frac{1}{R} + \frac{1}{\infty}\right] = \frac{(n-1)}{R} \Longrightarrow f = \frac{R}{(n-1)}$$

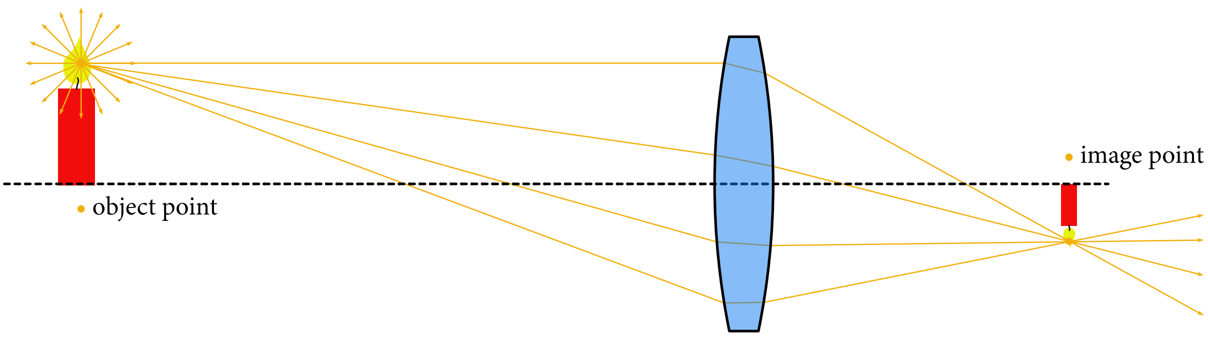
- Symmetrical biconcave lens ($R_1 = R_2 = -R$ with $R > 0$)

  $$\frac{1}{f} \approx (n-1)\left[\frac{1}{-R} + \frac{1}{-R}\right] = -\frac{2(n-1)}{R} \Longrightarrow f = -\frac{R}{2(n-1)}$$

Image Formation

Converging lenses are often used to create an image of light emitted by or reflected on an object. Typical applications are cameras or telescopes.

For a sharp image, divergent light rays emitted by a point of the object have to be deflected through the lens into an image point.



Remarks:

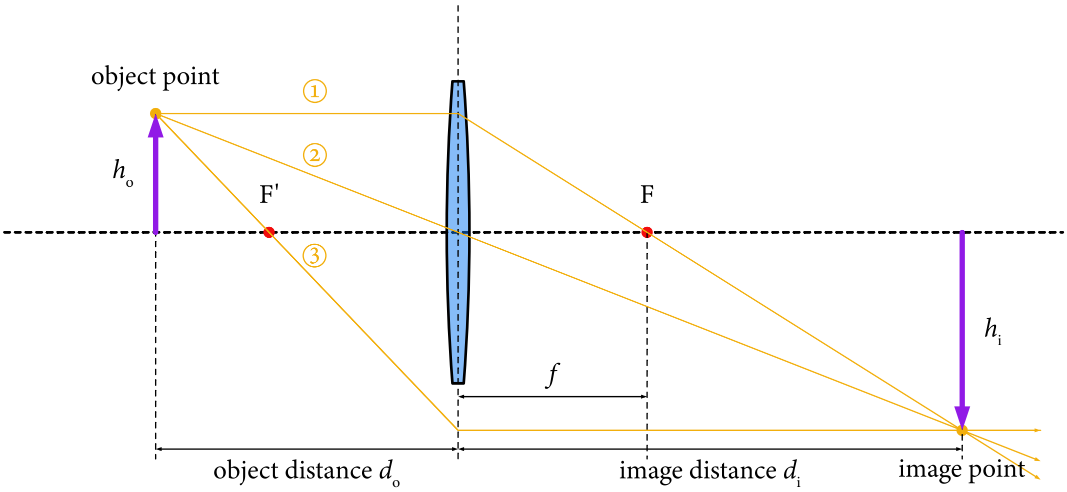
- Each object point has its respective image point. The image is the collection of all image points.
- For spherical lenses, the rays emanating from an object point are only approximately deflected through an image point.

Lens Equation

For thin lenses, we can approximate that the two refractions occur at the same point (the principal plane) and that a ray passing through the center of the lens is not deflected. We can then predict the deflection of three specific rays originating from an object point:

1. The ray parallel to the optical axis is deflected through the focal point F on the image side of the lens.
2. The ray passing through the centre of the lens is not deflected.
3. The ray passing through the focal point F' on the object side of the lens is deflected to become parallel to the optical axis.

Every other ray passing through the lens is deflected through the intersection of these three rays, so this is the image point.



Using similar triangles in the figure above, we find the following relations:

$$\frac{h_i}{h_o} = \frac{d_i}{d_o} \quad \text{and} \quad \frac{h_i}{h_o} = \frac{d_i - f}{f}$$

It follows that

$$\frac{d_i}{d_o} = \frac{d_i - f}{f}$$

Dividing both sides by d_i yields

$$\frac{1}{d_o} = \frac{d_i - f}{d_i f} = \frac{1}{f} - \frac{1}{d_i}$$

This is the thin lens equation, which in its standard form is usually written as

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

Lateral Magnification

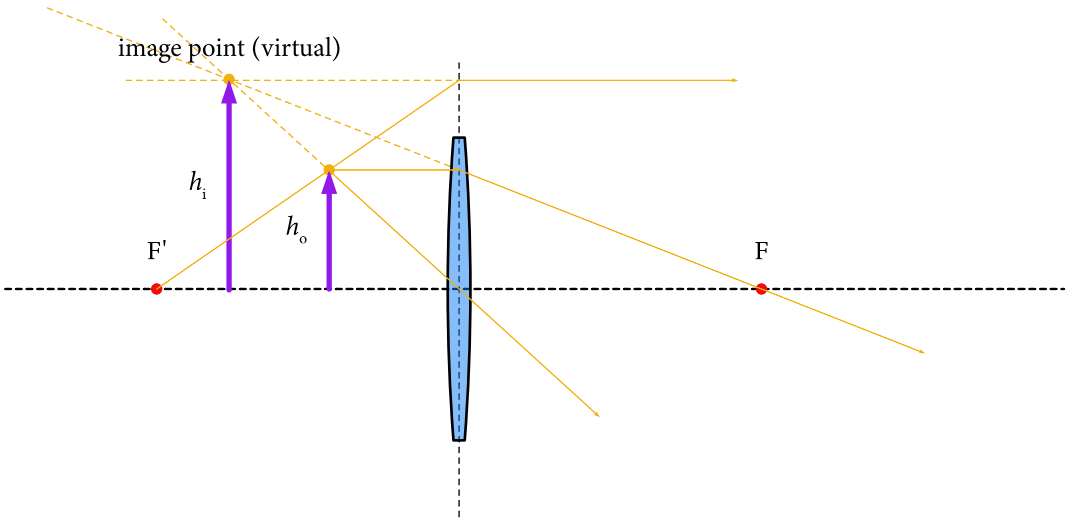
In general, the image has a different size than the object. The *lateral magnification* is given by

$$m = \frac{h_i}{h_o} = -\frac{d_i}{d_o}$$

where a negative sign means that the image is reversed (upside down).

Remarks:

- For $d_o > 2f$, the image is smaller than the object ($m < 1$).
- For $2f > d_o > f$, the image is larger than the object ($m > 1$).
- For $d_o <$ f, the image distance becomes negative. The reason is that the refracted light rays are divergent, but they appear to emanate from a point in front of the lens. This is a case where a *virtual image* can be observed.

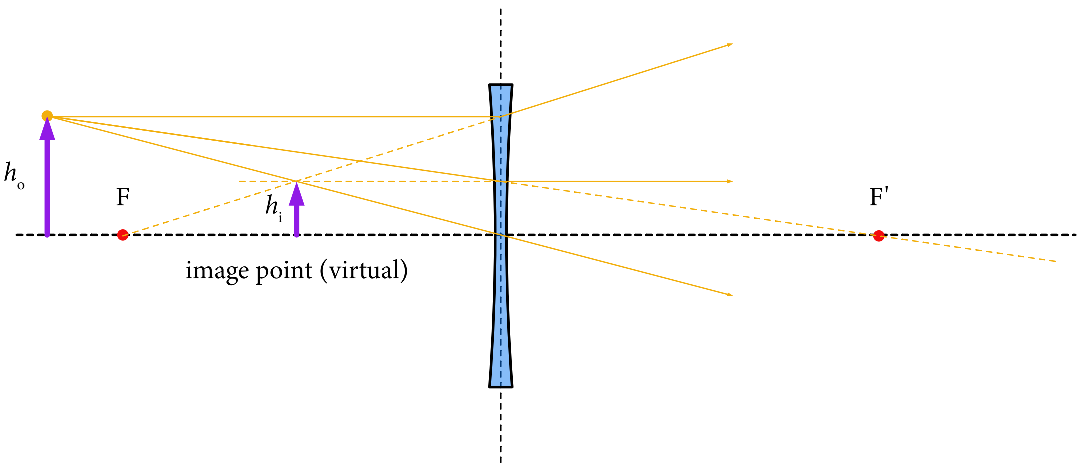


Images with Diverging Lenses

The lens equation can be applied to diverging lenses. In this case, the focal length has a negative value ($f < 0$). As a consequence, diverging lenses can only produce virtual images (unless the object distances is negative, e.g. for systems of lenses).

Example: Virtual image with diverging lens TEX

An object is placed 60 cm from a diverging lens with focal length – 50 cm. Determine the image.



The image distance is

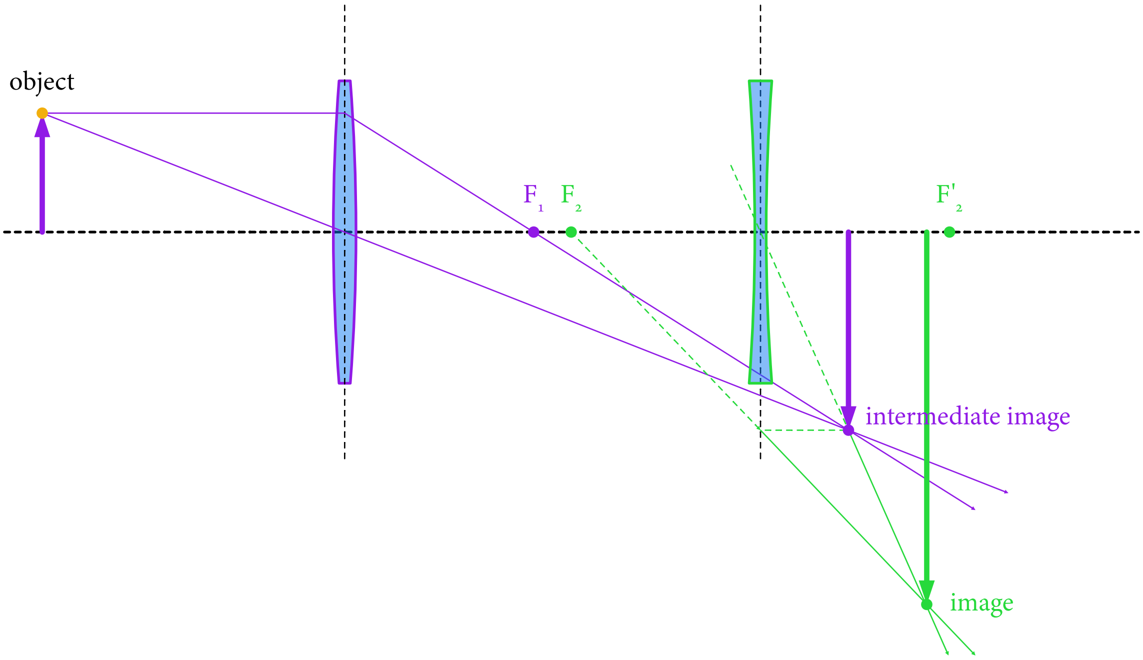
$$d_i = \left(\frac{1}{f} - \frac{1}{d_o}\right)^{-1} = \left(\frac{1}{-50\text{ cm}} - \frac{1}{60\text{ cm}}\right)^{-1} = -27\text{ cm}$$

Systems of Lenses

The image for a system of lenses can be determined step by step: The image distance of the image distance defines the object distance to the second lens, etc. If the image is after a lens, the corresponding object distance has a negative sign.

Example: System with a converging and a diverging lens [TeX]

In the system below, an object is placed 40 cm before a converging lens with focal length f_1 = 25 cm. The diverging lens with focal length f_2 = – 25 cm is placed 55 cm after the first lens. Determine the image.



The position of the intermediate image is given by

$$\frac{1}{f_1} = \frac{1}{d_{o,1}} + \frac{1}{d_{i,1}} \quad\Longrightarrow\quad d_{i,1} = \left(\frac{1}{f_1} - \frac{1}{d_{o,1}}\right)^{-1}$$

The numerical result is $d_{i,1}$ = 200/3 cm. The distance to the second lens is therefore $d_{o,2}$ = (55 – 200/3) cm. It follows for the position of the final image

$$d_{i,2} = \left(\frac{1}{f_2} - \frac{1}{d_{o,2}}\right)^{-1}$$

with a numerical value $d_{i,2}$ = 21.875 cm = 22 cm from the second lens.

The lateral magnification corresponds to the product of the magnifications for the first and second lens:

$$m = \frac{d_{i,1}}{d_{0,1}} \cdot \frac{d_{i,2}}{d_{o,2}}$$

with a numerical value m = – 3.125 = – 3.1. The image is about three times larger than the object and upside down, as can be verified in the construction.